Day 6 - Oct 25: Weekly Test, Business Questions, How LLMs Work

Swagat | Revision sheet | Open: AI Skill Hour (Oct 27 afternoon)

1. KEY IDEAS

Weekly test: Python

- FizzBuzz order: multiple of both 3 and 5 first, then 3, then 5, else the number. Sum only the numbers that were not multiples: put total += i in the else. Print the words exactly, with no extra space.
- 7 // 2 is an int (3). 7 / 2 is 3.5. "7" + 3 raises TypeError: can only concatenate str (not "int") to str.

Weekly test: Excel

- SUMIFS(sum range, check range, condition, ...): sum range first. COUNTIFS(range, "condition", range, "condition"): pairs of range and condition, the operator goes inside the quotes (">50000"), commas between all parts. N conditions need 2N-1 commas.
- A test like B2="Laptop" belongs inside IF, AND, OR. COUNTIFS and SUMIFS take a range plus a text condition.
- IF labels print exactly as typed: "Bulk" is not " Bulk". Grid: =$D2*G$1, the $ goes on the part that stays.

Weekly test: Git first push

```
!git config --global user.name "Name"
!git config --global user.email "you@mail.com"
!mkdir git-practice
%cd git-practice
!git init
!echo "print('Who')" >> example.py
!git add example.py
!git commit -m "First commit"
!git branch -M main
!git remote add origin https://github.com/USER/REPO.git
!git push -u origin main
```

- git push origin fails without remote add: origin is only a nickname for the GitHub URL, and Git has no remote with that name until you add it.

Business topic: from 'sales are falling' to questions

- The sentence is not investigable. Ask: which products, where (region/city), since when, compared with what (last month, same month last year, target).
- Revenue = orders x items per order x average price per item. A drop can come from any one factor, including a mix shift to cheaper products.
- Data: orders.order_date and order_id for orders; order_items.quantity; products.price; customers.city through joins. External: festival calendar, list of products being phased out.
- Investigation order: confirm the drop is real (complete month, days in month, same month last year) -> find when and where -> split into the three factors -> ask for external data -> recommend as if/then hypotheses.

AI Fluency

- An LLM predicts the next token. It does not look facts up, so it can hallucinate: confident, fluent, wrong. Specific details (numbers, titles, links) are the usual victims.
- It does not calculate. For a total use SUM in Excel or sum() in Python. Its knowledge stops at a cutoff date. Web search helps, but read the official page for anything important.
- LLM = the engine (the model). Chatbot = the product around it (chat window, web search, code tool, memory).
- Verify with evidence: run the code, read the docs. Colab keeps variables between cells, so use Restart and run all before trusting a notebook.

2. MY MISTAKES AND THE FIX

- FizzBuzz printed " FizzBuzz" with a leading space -> strings print exactly as typed.
- "7" + 3: gave the reason but not the error name -> TypeError.
- SUMIFS formula was right but I gave no value (260000) -> always give the formula AND the value.
- COUNTIFS: wrote =COUNTIFS(B2="Laptop",D2>50000), then forgot the = and a comma. Fix: =COUNTIFS(B2:B9,"Laptop",D2:D9,">50000") = 4 (rows 2, 4, 5, 9).
- IF labels typed with leading spaces (" Bulk", " -") -> exact text only.
- Git: listed names instead of commands; garbled 'git orinin M-main'; forgot git remote add origin. Later: no space in user.name"x", an email with no @, !%cd instead of %cd, file named www.com.
- Business: called 'which month revenue dropped' external -> it is internal (Q4 query). City needs a join chain, price needs the products table.
- AI: said an LLM 'uses the data it was trained on' -> it predicts the next token. Said numbers 'get mixed up' -> it does not calculate at all. Two reasons for stale answers were one reason: the knowledge cutoff.
- Verification: wrote 'you were right' with no evidence. The run showed the AI was partly right: its NameError claim failed because Colab had kept count from earlier cells.
- Recall log: wrote 'FizzBuzz space was not there' (there was an extra space) and 'performed good in Excel' (COUNTIFS had to be repeated).

3. HOW I IMPROVED

- Repeated COUNTIFS from scratch: right shape (range, condition), right value, right rows.
- Rewrote the Git push sequence with all steps in order, including remote add.

- Wrote a 5-line investigation with confirm, locate, split, external data and a conditional recommendation.
- Tested an AI answer in Colab and reported what each case printed, including the error it got wrong.

4. PRACTICE TO RETRY (no solutions)

- FizzBuzz plus sum for n = 15 and n = 0. Predict type(7 // 2), 7 / 2 and "7" + 3 without running.
- COUNTIFS with two conditions, SUMIFS with two, IF(AND(...)) label, price x rate grid, all from a blank sheet.
- From an empty folder: identity, init, file, add, commit, rename branch, add remote, push.
- Write the 5-line investigation for 'orders fell 15%' from memory.

5. RECALL QUESTIONS (answer without notes)

- What is the shape of a COUNTIFS argument? Why can B2="Laptop" not go in it?
- Why does git push origin fail before git remote add origin?
- Name the four questions that turn 'sales are falling' into an investigation.
- Write revenue as a product of three factors.
- Why can an LLM be confidently wrong? Why use SUM instead of asking it for a total?
- LLM vs chatbot in one line each.

6. OPEN ITEMS

- AI Skill Hour (20 min reading, 3 lines): Oct 27 afternoon. Day 6 README commit: pending.